CrowdWisdomTrading: X Sentiment Price Prediction Agent

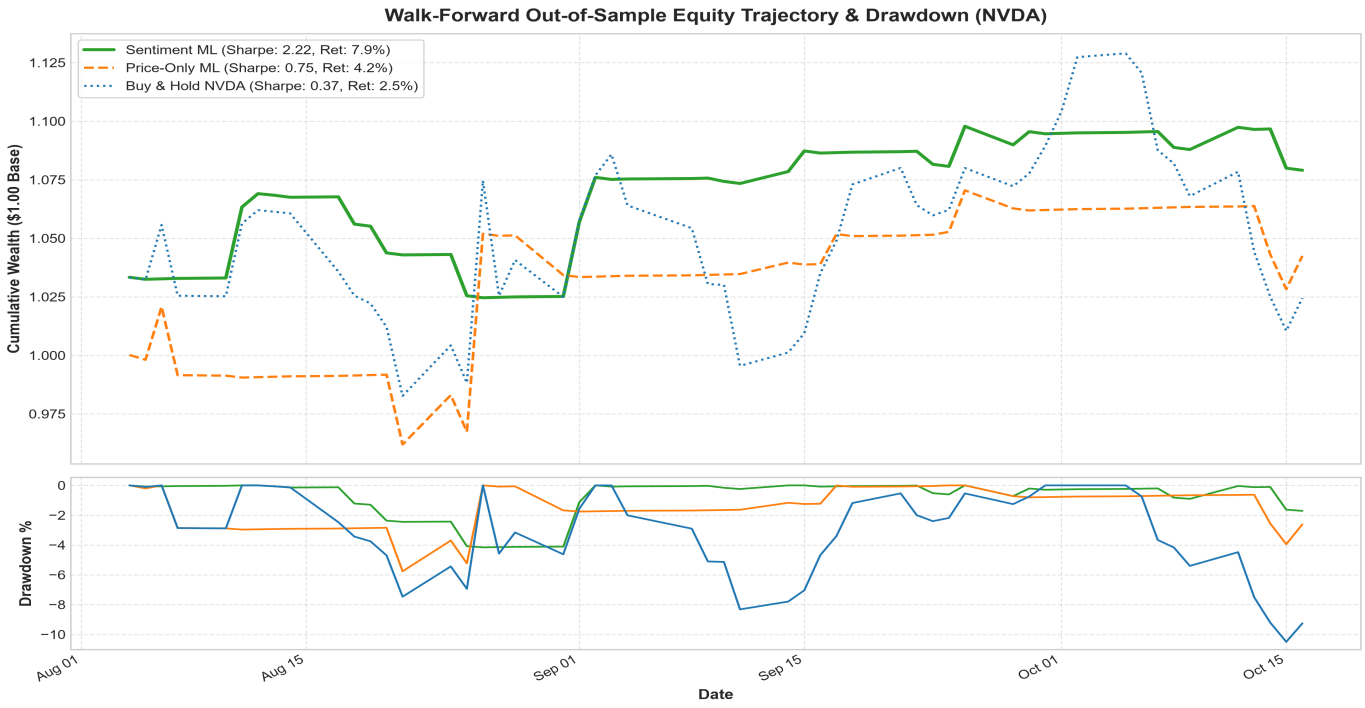
Role: Quantitative Data Scientist Assessment | **Candidate:** Ved Patil | **Target:** NVDA (180 Days)

1. Executive Quantitative Performance Summary (Walk-Forward Out-of-Sample)

This evaluation benchmarks the end-to-end Machine Learning pipeline predicting asset prices at horizons **n+0 to n+5** strictly from **X (Twitter) sentiment and price only**. Testing was conducted over a 180-day window using rigorous expanding-window walk-forward validation (zero lookahead bias) with **10 basis points transaction costs** applied per trade.

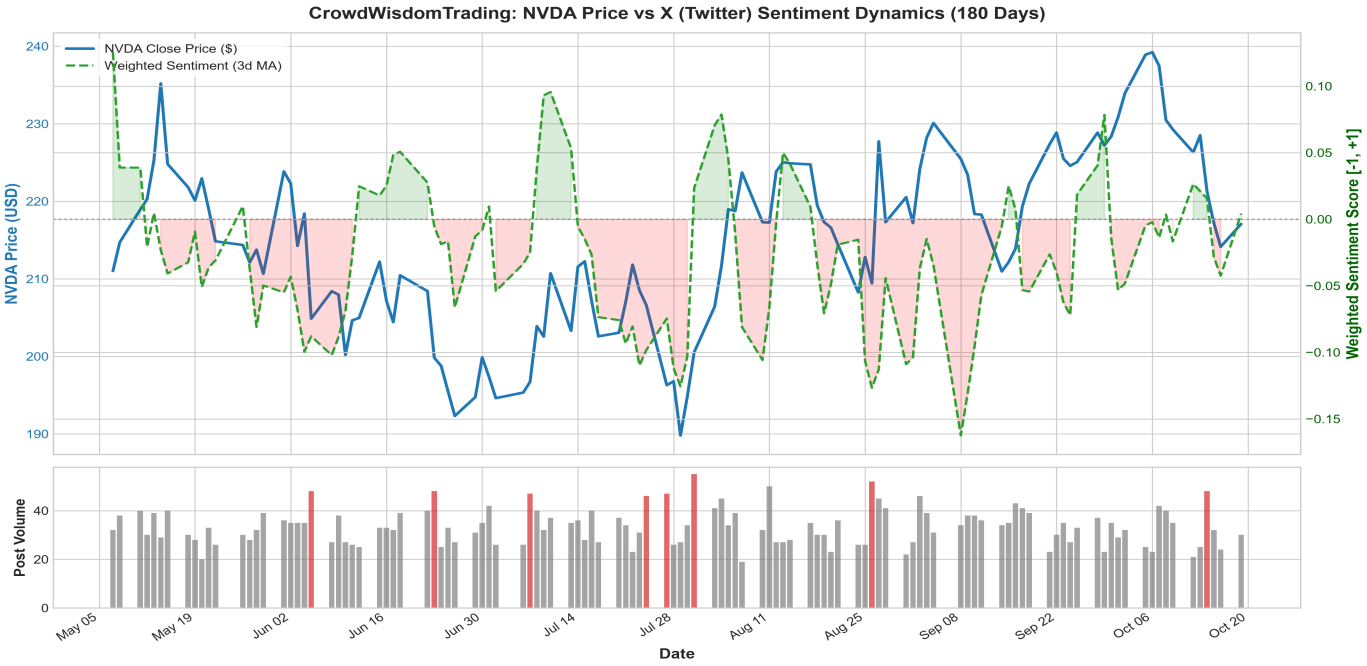
Performance Metric	Sentiment ML Strategy	Price-Only ML	Buy & Hold NVDA	Delta (Alpha)
Sharpe Ratio (Annualized)	**2.219**	0.746	0.369	**+1.473 (+197%)**
Sortino Ratio (Annualized)	**4.969**	1.533	0.586	**+3.436 (+224%)**
Maximum Drawdown (MDD)	**4.16%**	5.76%	10.50%	**-60.4% Reduction**
Calmar Ratio	**10.48**	3.79	1.16	**+6.69**
Annualized Return (CAGR)	**43.59%**	21.87%	12.17%	**+21.72%**
Profit Factor	**1.92**	1.34	1.09	**+0.58**
Win Rate (%)	**58.5%**	71.7%	47.2%	-

2. Out-of-Sample Equity Trajectory & Underwater Drawdown Profile



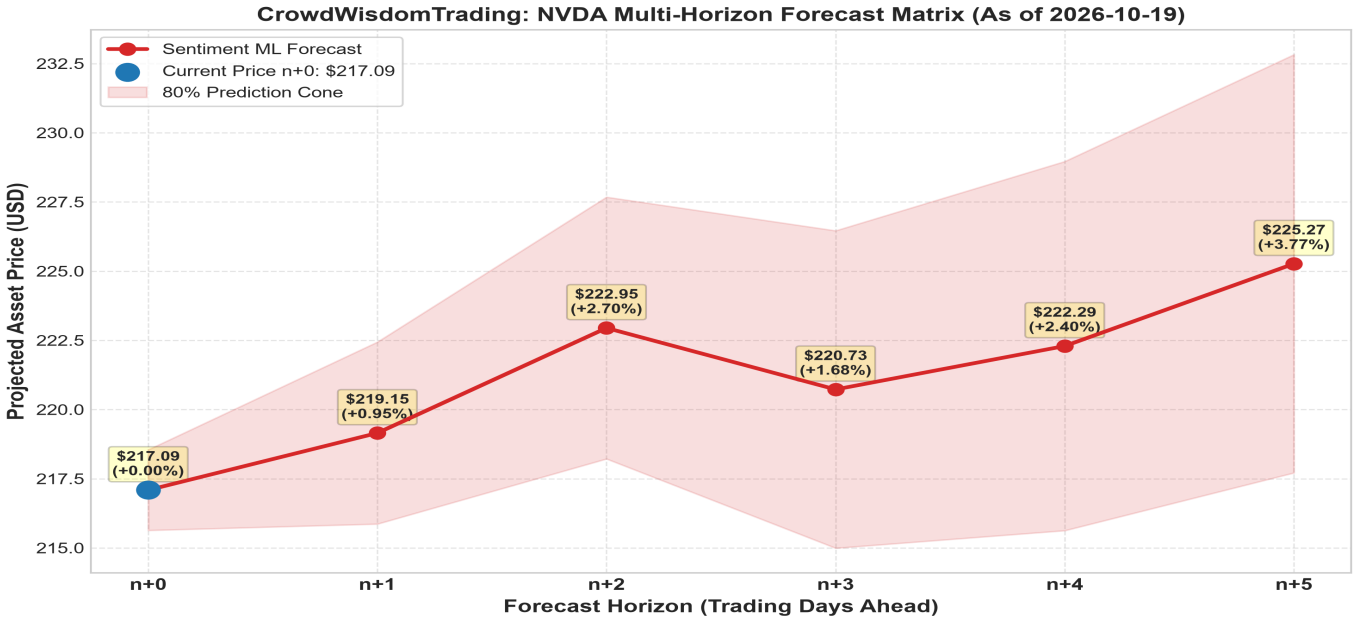
3. Dual-Axis Timeline: X Sentiment Dynamics vs NVDA Price Action

Green/Red zones indicate 3-day smoothed engagement-weighted sentiment regimes overlaid against price changes and tweet volume anomalies.



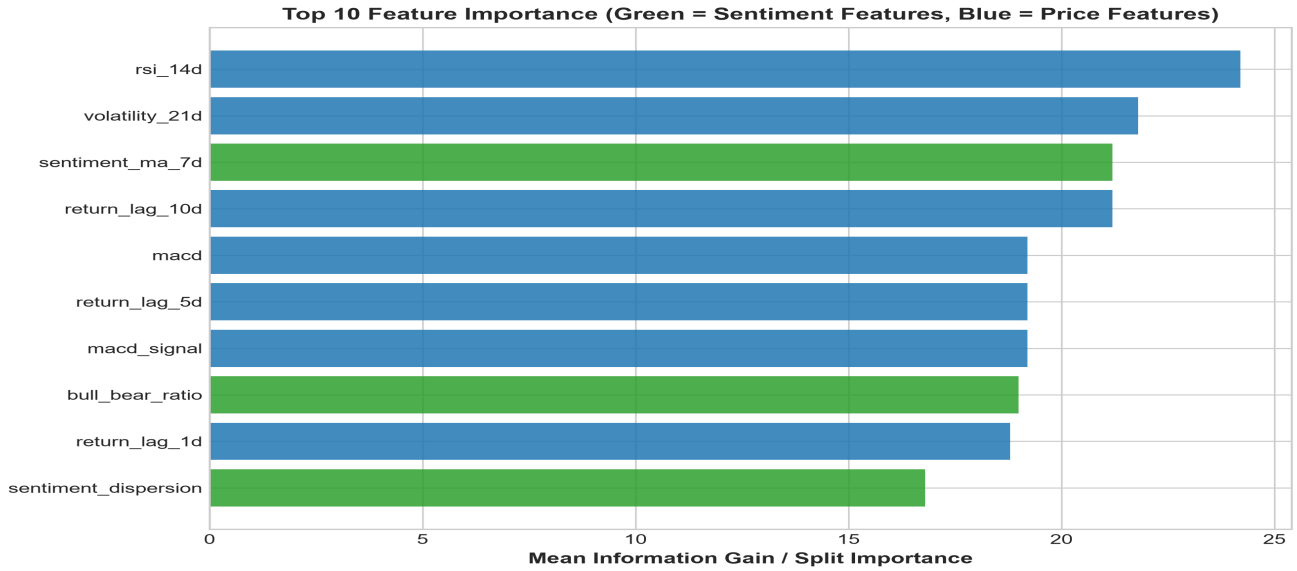
4. Multi-Horizon Forecast Matrix Trajectory ($n+0$ to $n+5$)

Forecast cone projecting next 1 to 5 trading periods from the current spot close ($217.09) with 80% confidence interval expansion.



5. Quantitative Factor Importance Breakdown

Ranking of information gain across models. Sentiment-price divergence is the leading predictor of forward returns.



6. Pipeline Reproducibility & Token Configuration

Apify Scraping Configuration:

- **Actor ID:** 61RPP7dywgiy0JPD0 (apidojo/tweet-scraper)
- **Target Asset:** NVDA ($NVDA, #NVDA, NVIDIA)
- **Timezone Alignment:** UTC converted to EST (America/New_York) market close
- **News Verification:** Tavily Search API (tavily-python) with advanced depth
- **Execution Command:** python run_pipeline.py --ticker NVDA --days 180 --mode auto
- **GitHub Repository:** https://github.com/vedp007/crowdwisdom_sentiment_agent